Reports Feature Roadmap — Business IQ Enterprise

Enterprise-grade report generation with AI-powered content, configurable formatting, and professional export quality.

Phase 1: Foundation ☑ (Complete)

1.1 Core Report Pipeline

- ☑ Report Generator Agent (LLM writes professional reports from SQL data)
- ☑ Fast path (pre-computed data) and SQL path (dynamic queries)
- ☑ Multi-format export: PDF, XLSX, DOCX, PPTX
- ☑ S3 storage with presigned download URLs
- ☑ Temp cache fallback when S3 is unavailable
- ☑ Report caching (KV cache with TTL)
- ☑ Report persistence (saved to database with versioning)

1.2 PDF Template

- ☑ Branded title page (company logo, name, tagline, report title)
- ☑ Table of Contents page
- ☑ Section headings with brand color accent lines
- ☑ Elegant table rendering (alternating stripes, word wrap, column auto-sizing)
- ☑ Vertical column separators and outer frame borders
- ☑ Multi-page table support with header re-draw
- ☑ Per-page border segment tracking (correct multi-page rendering)
- ☑ Page footers (company name, page numbers)

1.3 LLM Prompt Engineering

- ☑ Structured report format (Executive Summary → Key Metrics → Analysis → Rankings → Conclusion → References)
- ☑ No-placeholder enforcement (company name, dates injected directly)
- ☑ PII masking on output
- ☑ Custom instructions from admin console
- ☑ Read-only SQL guardrails

Phase 2: Visual Intelligence & Configuration ☑ (Complete)

2.1 Chart Generation Pipeline

- ☑ Vega-Lite + Vega + Sharp (SVG → PNG at 2x resolution)
- ☑ Support: bar, line, area, pie, donut, scatter, grouped_bar, stacked_bar, heatmap
- ☑ Brand-aware color palette (primary color → complementary series)
- ☑ LLM generates ```chart JSON blocks inline with report content

- ☑ `extractChartBlocks()` parses and validates chart specs from markdown
- ☑ Charts embedded in PDF with titles, accent underlines, bordered frames, figure captions
- ☑ Charts embedded in XLSX (as PNG images), DOCX (ImageRun), PPTX (native charts + PNG fallback)
- ☑ `toPptxChartData()` for native PowerPoint chart rendering

2.2 PDF Formatting Polish

- ☑ Table titles (bold, brand color) above tables
- ☑ Numbered "Table N:" captions (italic, centered) below tables
- ☑ Chart titles with accent underline above charts
- ☑ Numbered "Figure N:" captions (italic, centered) below charts
- ☑ References section with special formatting (heading, accent line, numbered items, hanging indent)
- ☑ Trailing empty page removal
- ☑ Smart table space estimation (prevents orphaned titles)
- ☑ LLM metadata stripping (removes duplicate title, "Prepared for:", "Date:", "---" from content pages)

2.3 Admin Console — Report Settings

- ☑ "Reports" tab under Operations section in Admin Console
- ☑ Toggle: Title Page on/off
- ☑ Toggle: Table of Contents on/off
- ☑ Toggle: References Section on/off
- ☑ Toggle: Confidential Footer on/off
- ☑ Toggle: Charts Enabled on/off
- ☑ Number: Executive Summary word target (default: 200)
- ☑ Number: Max page count (default: 20)
- ☑ Number: Max word count (default: 5000)
- ☑ Number: Max charts per report (default: 4)
- ☑ Number: Max data points per chart (default: 15)
- ☑ Settings persisted to DB, cached with 1-min TTL
- ☑ Cache invalidation on save

2.4 Bug Fixes

- ☑ Company name reads from DB `businessName` (not env var fallback)
- ☑ Duplicate footer on title page removed (title page has its own "Confidential" footer)
- ☑ Executive Summary continues on TOC page if room (>250px) instead of starting new page
- ☑ Multi-table extraction (`extractTables` replaces `extractTable`)

Phase 3: Advanced Formatting (Next)

3.1 Watermark & Security

- ☐ Configurable watermark text ("DRAFT", "CONFIDENTIAL", custom) — diagonal, semi-transparent
- ☐ Watermark on/off toggle in report settings
- ☐ Document password protection (PDF encryption)
- ☐ Digital signature embedding

3.2 Header & Footer Customization

- ☐ Configurable header content (logo position, company name, report title)
- ☐ Configurable footer template (company name, page numbers, date, custom text)
- ☐ Different first-page header/footer vs subsequent pages
- ☐ Section-specific headers (show current section name)

3.3 Typography & Styling

- ☐ Custom font support (embed TTF/OTF fonts)
- ☐ Font family selection in report settings (Serif, Sans-serif, Monospace)
- ☐ Custom brand color palette (primary, secondary, accent)
- ☐ Table style presets (minimal, bordered, striped, modern)
- ☐ Configurable page margins
- ☐ Line spacing control (single, 1.5, double)

3.4 Rich Content

- ☐ Callout boxes / highlighted insight panels ("Key Insight:", "Warning:", "Recommendation:")
- ☐ Key metric cards (large numbers with trend indicators ↑↓)
- ☐ Inline data sparklines (miniature charts in text)
- ☐ Footnotes with auto-numbering
- ☐ Cross-references between sections ("See Section 3.2")

Phase 4: AI-Powered Visuals

4.1 AI Image Generation

- ☐ Integration with image generation API (DALL-E 3, Stable Diffusion, or Flux)
- ☐ Industry-relevant decorative images (e.g., product photos, business context imagery)
- ☐ Auto-generated section header illustrations
- ☐ Image placement settings (inline, sidebar, full-width)
- ☐ Image generation on/off toggle in report settings
- ☐ Image style presets (photorealistic, illustration, infographic)
- ☐ Cost control: max images per report setting

4.2 Advanced Charts

- ☐ Combination charts (bar + line overlay)
- ☐ Waterfall charts (period-over-period changes)
- ☐ Gauge charts (KPI vs target visualization)
- ☐ Funnel charts (sales pipeline, conversion rates)
- ☐ Treemap charts (hierarchical data — category > product revenue)
- ☐ Geographic/map charts (regional sales heatmap)
- ☐ Sparkline mini-charts embedded in table cells
- ☐ Interactive HTML chart export (for web/email reports)

4.3 Chart Intelligence

- ☐ Auto-chart selection (LLM picks optimal chart type for each dataset)
- ☐ Trend annotations (auto-add trend lines, highlight anomalies)
- ☐ Comparison periods (this month vs last month overlay)
- ☐ Statistical annotations (mean line, standard deviation band)
- ☐ Chart color legend improvements (inline labels instead of separate legend)

Phase 5: Report Templates & Scheduling

5.1 Template Engine

- ☐ Pre-built report templates (Sales Summary, Inventory Health, Financial Overview, Customer Activity)
- ☐ Custom template builder (drag-and-drop sections)
- ☐ Template versioning (save iterations, rollback)
- ☐ Template sharing across deployments (export/import)
- ☐ Section library (reusable content blocks)
- ☐ Conditional sections (include/exclude based on data availability)

5.2 Scheduled Reports

- ☐ Recurring report generation (daily, weekly, monthly, quarterly, annual)
- ☐ Report calendar view in admin console
- ☐ Email delivery of generated reports (PDF attachment or download link)
- ☐ Distribution lists (send to multiple recipients)
- ☐ Report queue management (cancel, retry, priority)
- ☐ Notification on report completion (in-app + email)

5.3 Multi-Period Reports

- ☐ Comparison reports (this period vs last period, YoY)
- ☐ Rolling period reports (last 7 days, 30 days, 90 days, 12 months)
- ☐ Custom date range picker with presets
- ☐ Period-over-period trend analysis with % change
- ☐ Annual summary reports (fiscal year)

Phase 6: Collaboration & Review

6.1 Report Workflow

- ☐ Draft/review/approved/published status workflow
- ☐ Report approval chain (manager must approve before distribution)
- ☐ Annotation/commenting on report sections
- ☐ Version history with diff view
- ☐ Report locking (prevent edits after approval)
- ☐ Audit trail (who generated, reviewed, approved, shared)

6.2 Report Sharing

- ☐ Shareable links with expiry and access control

- [ ] Role-based report access (which roles can view which report types)
- [ ] White-label PDF export (remove software branding)
- [ ] Custom cover page per report (upload custom background image)
- [ ] QR code on cover page linking to online version

6.3 Report Library

- [ ] Searchable report archive (by type, date, generator, status)
- [ ] Report tagging and categorization
- [ ] Favorites/bookmarks
- [ ] Bulk download (zip multiple reports)
- [ ] Storage quota management

Phase 7: Smart Insights & Narrative Quality

7.1 Narrative Enhancement

- [ ] Executive tone calibration (formal, conversational, analytical)
- [ ] Audience-aware writing (C-suite summary vs operational detail)
- [ ] Multi-language report generation (translate to client's language)
- [ ] Industry-specific terminology injection (from config)
- [ ] Reading level adjustment (simple, standard, technical)

7.2 AI Analysis Depth

- [ ] Anomaly detection and auto-highlighting
- [ ] Root cause analysis for significant changes
- [ ] Predictive insights ("Based on trends, next month may see...")
- [ ] Competitive benchmarking (if external data available)
- [ ] Seasonality detection and commentary
- [ ] Actionable recommendations with priority scoring

7.3 Data Quality

- [ ] Data completeness scoring (% of period with records)
- [ ] Data confidence indicators (sample size, coverage notes)
- [ ] Missing data handling (explicit callouts vs imputation)
- [ ] Outlier detection and commentary

Phase 8: Multi-Format Excellence

8.1 PDF Enhancements

- [ ] Clickable hyperlinks (TOC entries link to sections)
- [ ] PDF bookmarks (navigation panel)
- [ ] Accessibility (tagged PDF, alt text for images/charts)
- [ ] Print-optimized layout (CMYK color, bleed marks)
- [ ] PDF/A compliance (archival format)

8.2 DOCX Enhancements

- ☐ Table of Contents with page numbers (using DOCX field codes)
- ☐ Track Changes metadata
- ☐ Custom styles matching PDF design
- ☐ Chart native embedding (OOXML charts)
- ☐ Template DOCX with pre-designed styles

8.3 PPTX Enhancements

- ☐ Speaker notes auto-generated per slide
- ☐ Slide master templates (branded theme)
- ☐ Executive summary slide (single-slide overview)
- ☐ One chart per slide layout
- ☐ Animation presets (build slides progressively)

8.4 New Formats

- ☐ HTML report (interactive, responsive, sharable web page)
- ☐ Email-ready HTML (inline styles, email-client compatible)
- ☐ Dashboard embed (iframe-ready report view)
- ☐ CSV/JSON export of underlying data tables
- ☐ Markdown export (clean, no chart blocks)

Phase 9: Enterprise Features

9.1 Compliance & Regulatory

- ☐ SOX compliance report templates
- ☐ GDPR data subject report generation
- ☐ Tax authority report formats (KRA, IRS, etc.)
- ☐ Custom regulatory template builder
- ☐ Automated compliance checklist in reports

9.2 Multi-Deployment Intelligence

- ☐ Cross-deployment analytics (aggregate across client deployments)
- ☐ Benchmark reports (compare client performance to anonymized averages)
- ☐ Industry benchmark data integration

9.3 Performance & Scale

- ☐ Background report generation (non-blocking)
- ☐ Report generation queue with priority levels
- ☐ Incremental report updates (append new data to existing report)
- ☐ Chart rendering cache (reuse identical charts across reports)
- ☐ CDN distribution for generated documents
- ☐ Report generation analytics (time, cost, tokens per report)

AI Image Generation — Provider Options

When image generation is prioritized, evaluate these options:

Provider	Model	Quality	Cost	Notes
OpenAI	DALL-E 3	High	$0.04-0.08/img	Best prompt adherence, already in stack
Stability AI	SDXL/SD3	High	$0.002-0.03/img	Self-hostable, lower cost
Black Forest Labs	FLUX.1	Very High	API pricing varies	Newest, excellent quality
Replicate	Various	Varies	Pay-per-use	Multiple model options

Recommendation: Start with OpenAI DALL-E 3 (already using OpenAI for LLM, simplest integration). Add a `reportImageProvider` setting in admin console to allow switching providers.

This roadmap is implementation-ordered. Each phase builds on the previous. Priorities may shift based on user feedback and business requirements.